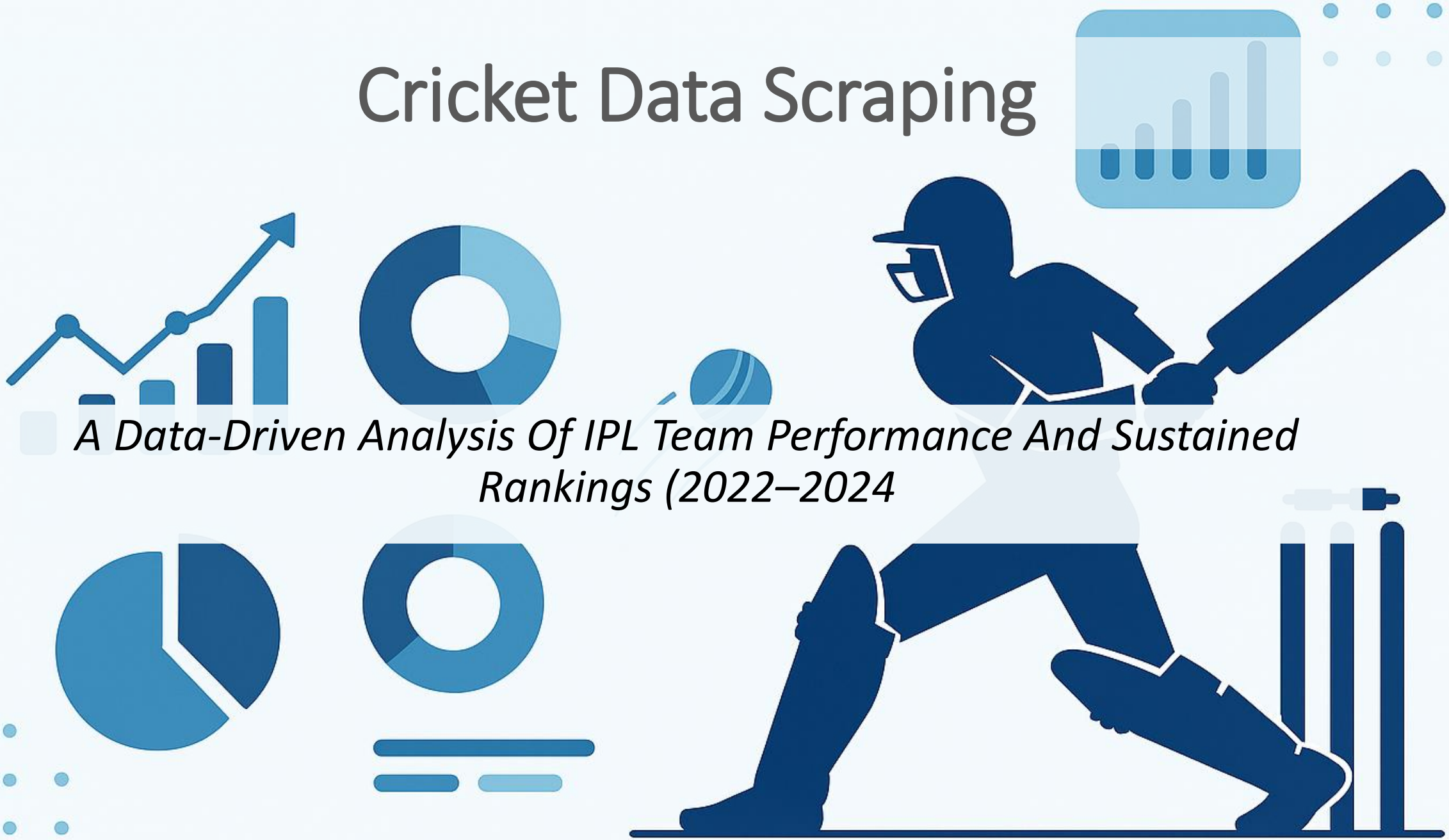
Cricket Data Scraping



A Data-Driven Analysis Of IPL Team Performance And Sustained Rankings (2022–2024

Introduction

- In most cricket tournaments, team performance is judged mainly by points and match wins. However, wins alone do not always show how strong or consistent a team actually is. A team may perform very well in one season and poorly in the next, or may depend heavily on only batting or bowling.

- This project aims to analyze IPL teams in a more meaningful way by using data analytics to study **performance quality, balance, and consistency** over three seasons.

Objective And Approach

- The primary aim of this study is to identify sustained performance rankings of IPL teams across the 2022, 2023, and 2024 seasons.

- Approach: Data collected from official IPL sources, cleaned, normalized, and analyzed across three seasons.

- Metrics: Win %, Net Run Rate (NRR), Batting/Bowling Strength, Balanced Performance

- In addition to conventional metrics, this study introduces two newly developed indices: the Season Dominance Index (SDI) and the Consistency Index.

Data Collection Process

Data Source : Data was collected from the **official IPL T20 website**

- Season-wise points table data for: 2022, 2023 and 2024.

- IPL points table data was scraped season-wise

- Team-level information was collected including: Matches played, Wins and losses, Net Run Rate, Runs scored and conceded

- **Tools Used :** Microsoft Excel, Chrome web scraping extension and Advance Excel operations

Data Cleaning and Pre-Processing

Raw data often contains inconsistencies. To fix this:

- Team names were standardized
- Net Run Rate formatting issues were corrected
- Missing values were handled
- Duplicate records were removed
- All numeric values were converted into proper formats
- After cleaning, the dataset became consistent and ready for analysis.

Team-Level Data Description

Original Column	Suggested Name	Description
POS	Rank	Final position of the team in the season points table
TEAM	Team Name	Name of the IPL team
P	Matches Played	Total number of matches played by the team
W	Wins	Total number of matches won
L	Losses	Total number of matches lost
NR	No Result	Number of matches with no result due to rain or abandonment
NRR	Net Run Rate (NRR)	Net run rate, indicating the scoring and conceding efficiency of the team
FOR	Total Runs Scored	Total runs scored by the team across the tournament
AGAINST	Total Runs Conceded	Total runs conceded by the team across the tournament
PTS	Points	Total points earned (2 points awarded per win)

Core Performance Metrics

Metric Name	What it Measures	Why it is Needed
Win Percentage	Match-winning efficiency	Allows fair comparison across teams regardless of matches played
Net Run Rate (NRR)	Run scoring and conceding efficiency	Shows dominance beyond wins and losses
Normalized NRR	Scaled NRR value (0–100)	Makes NRR comparable across seasons
Batting Runs per Over (RPO)	Batting efficiency per over	More meaningful than total runs scored
Bowling Runs per Over (RPO)	Bowling efficiency per over	Measures effectiveness in restricting runs
Balanced Performance Index (BPI)	Combined batting and bowling performance	Identifies well-balanced teams

Advanced Analytical Indices

Metric Name	What it Measures	Why it is Needed
Batting Strength Index	Normalized batting performance	Enables standardized comparison of batting quality
Bowling Strength Index	Normalized bowling performance	Rewards teams that concede fewer runs
Season Dominance Index (SDI)	Overall team dominance in a season	Combines results and performance quality
Consistency Index (CI)	Stability of performance across seasons	Identifies reliable long-term performers

Ranking Approach

Ranking Type	How the Ranking Was Done	Why This Ranking Was Needed
Season-wise Ranking	Teams were ranked separately for each season (2022, 2023, 2024) using performance metrics such as Win Percentage, NRR, Batting Strength, Bowling Strength, BPI, and SDI	To evaluate team performance within individual seasons and allow fair year-to-year comparison
Overall Rankings (2022 – 2024)	Performance metrics were averaged across all three seasons before ranking	To reduce the impact of one strong or weak season and measure long-term performance
Final Sustained Performance Ranking (2022 - 2024)	Teams were ranked using the **Average Season Dominance Index (SDI)** across 2022–2024	To identify teams that consistently performed well over multiple seasons
Consistency Ranking (2022 – 2024)	Teams were ranked based on the Consistency Index derived from SDI variation	To measure stability and reliability of team performance over time

Overall Rankings (2022 – 2024)

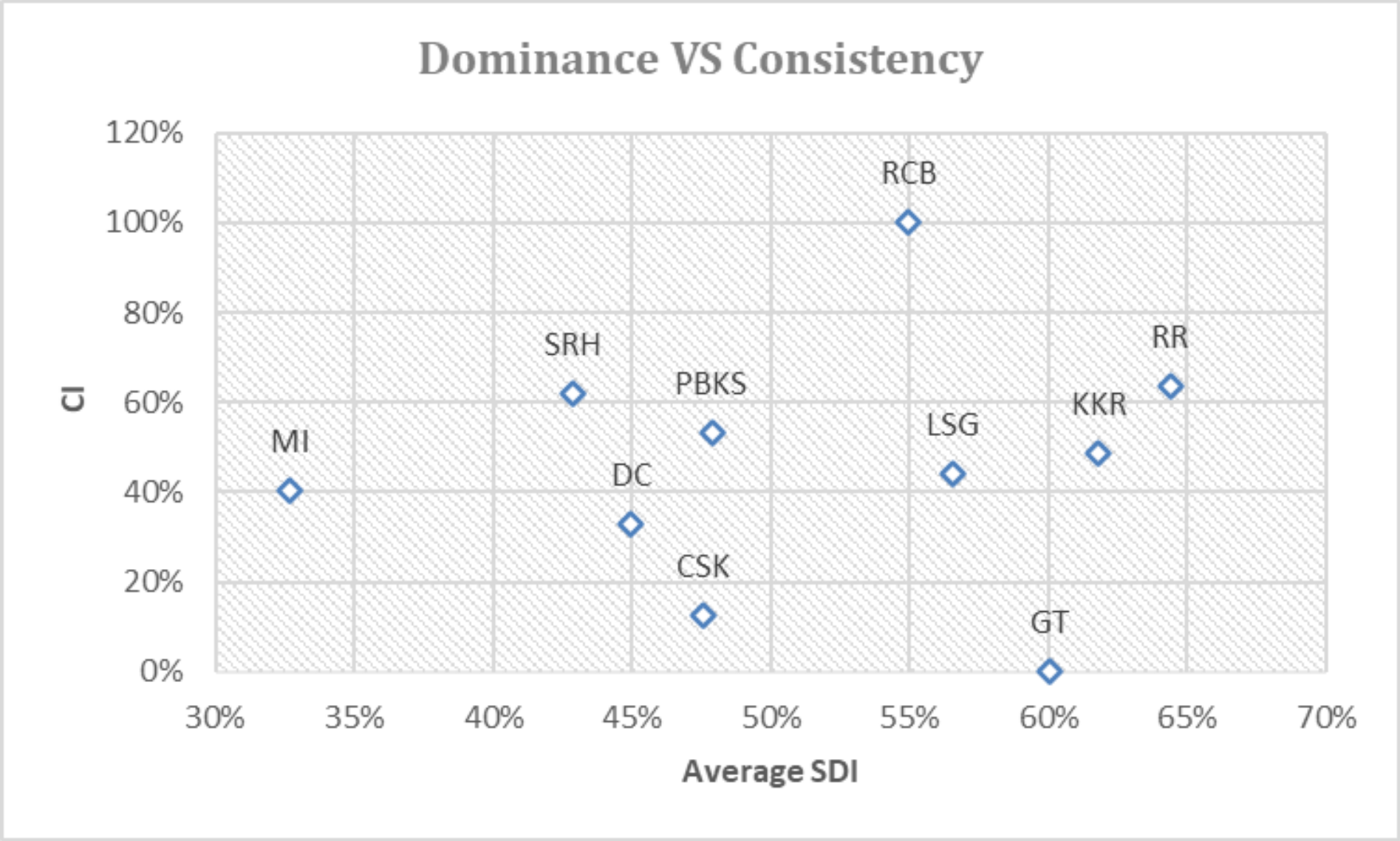
SL. NO	Teams	Win % Rank	NRR Rank	Batting Strength Rank	Bowling Strength Rank	Balanced Performance	Overall Performance Rank	Consistancy Rank
1	GT	1	3	10	1	3	3	10
2	RR	3	2	2	2	1	1	2
3	LSG	2	5	9	3	4	4	6
4	RCB	4	4	4	6	5	5	1
5	DC	6	9	6	7	8	8	8
6	PBKS	8	7	3	8	6	6	4
7	KKR	5	1	1	4	2	2	5
8	SRH	9	8	5	9	9	9	3
9	CSK	7	6	8	5	7	7	9
10	MI	10	10	7	10	10	10	7

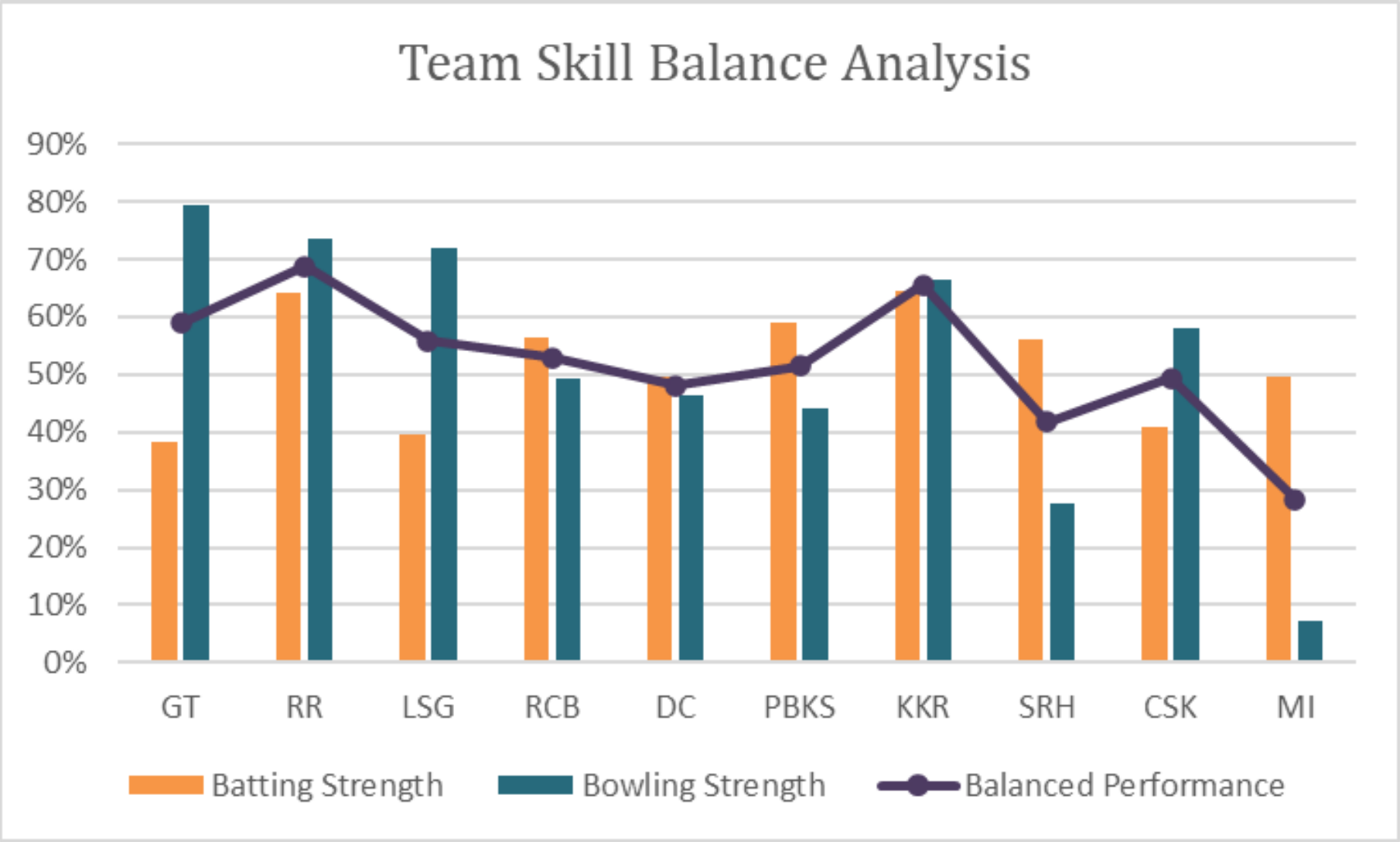
Final Sustained Performance Rankings (2022–2024)

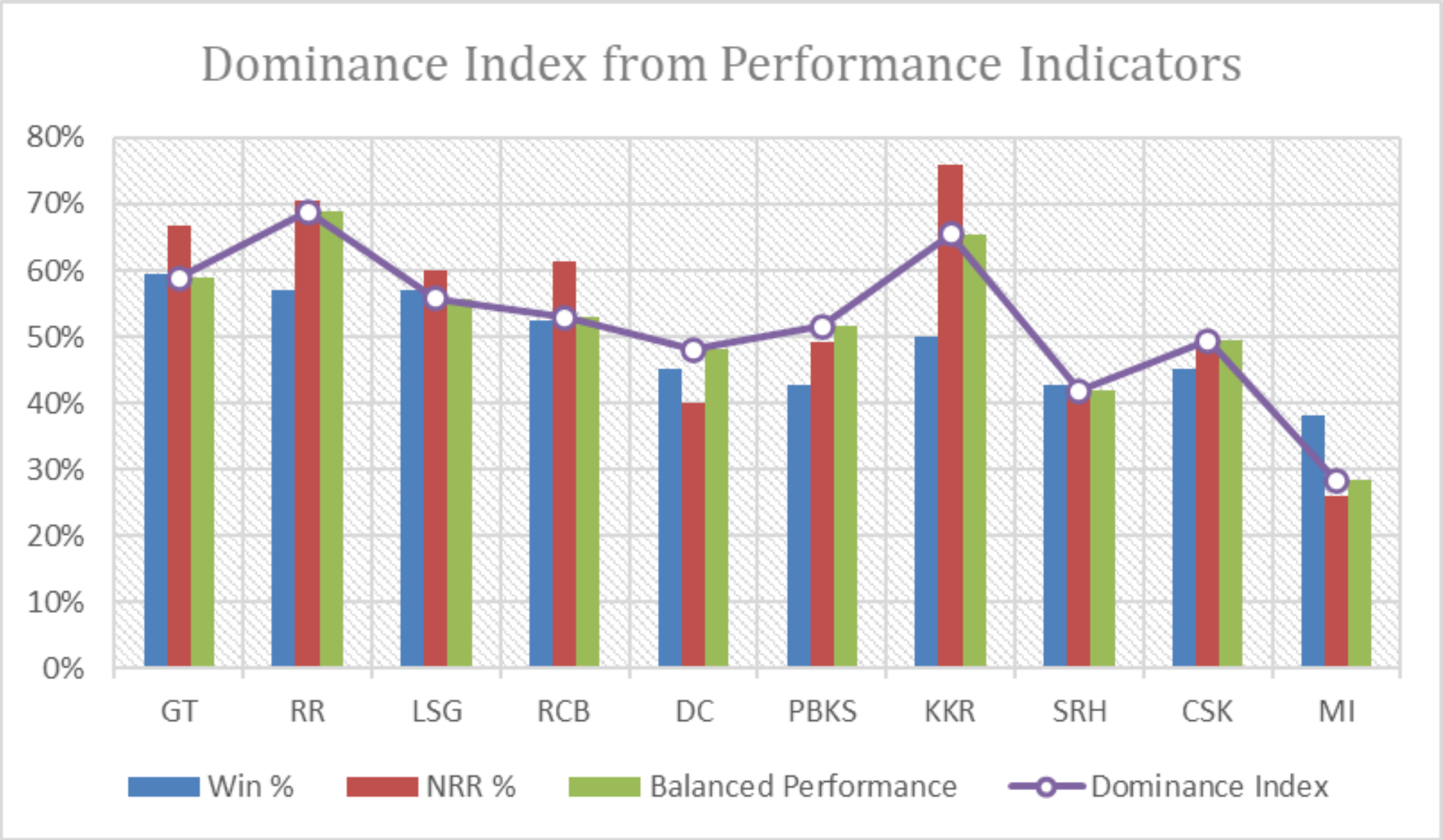
Rank	Team	Why They Rank Here
1	Rajasthan Royals (RR)	Highest average SDI with strong balance and consistency
2	Kolkata Knight Riders (KKR)	Consistently high dominance across all three seasons
3	Gujarat Titans (GT)	Strong overall dominance, mainly bowling-driven
4	Lucknow Super Giants (LSG)	Stable and well-balanced performance over time
5	Royal Challengers Bengaluru (RCB)	Moderate dominance with very high consistency

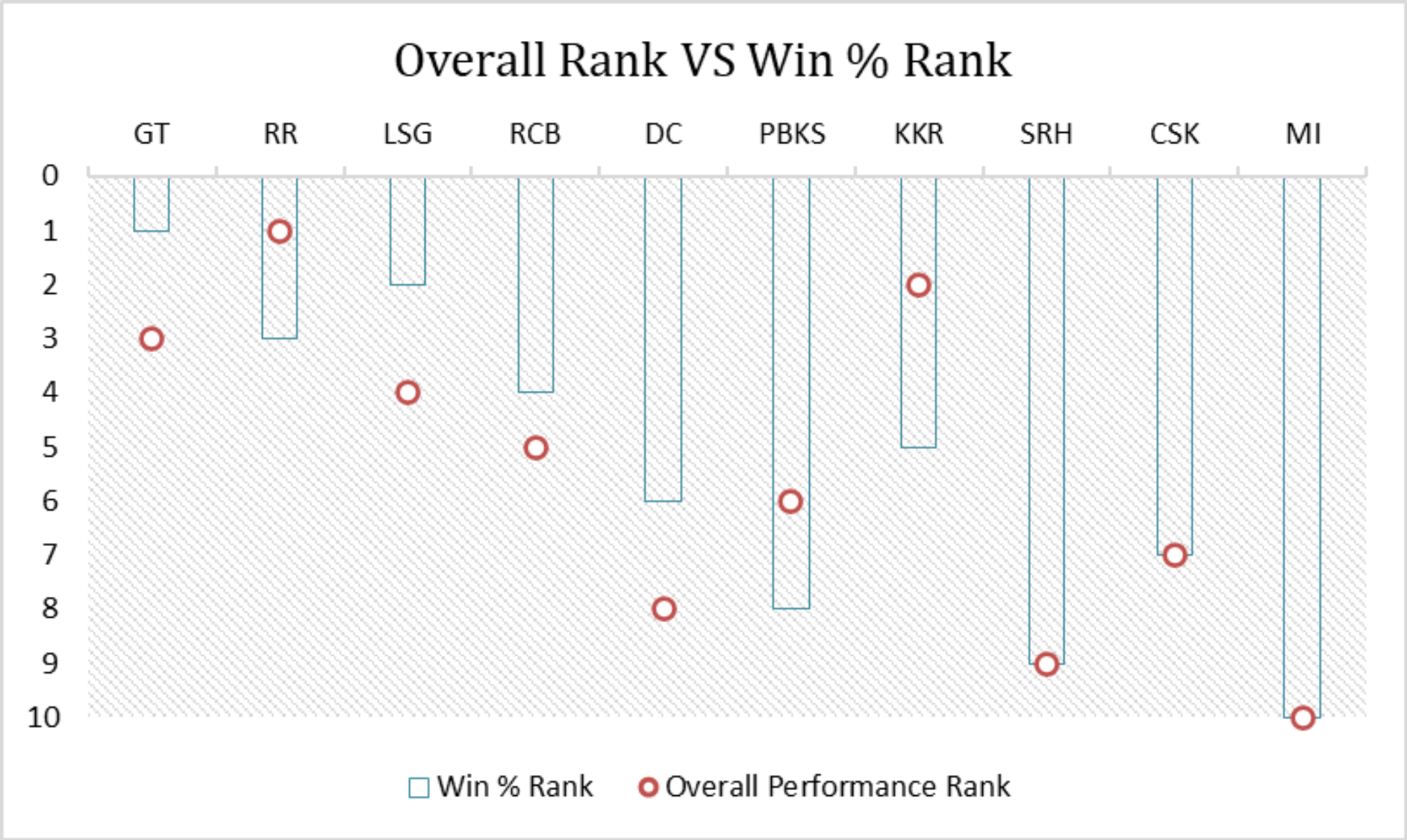
Final Sustained Performance Rankings (2022–2024)

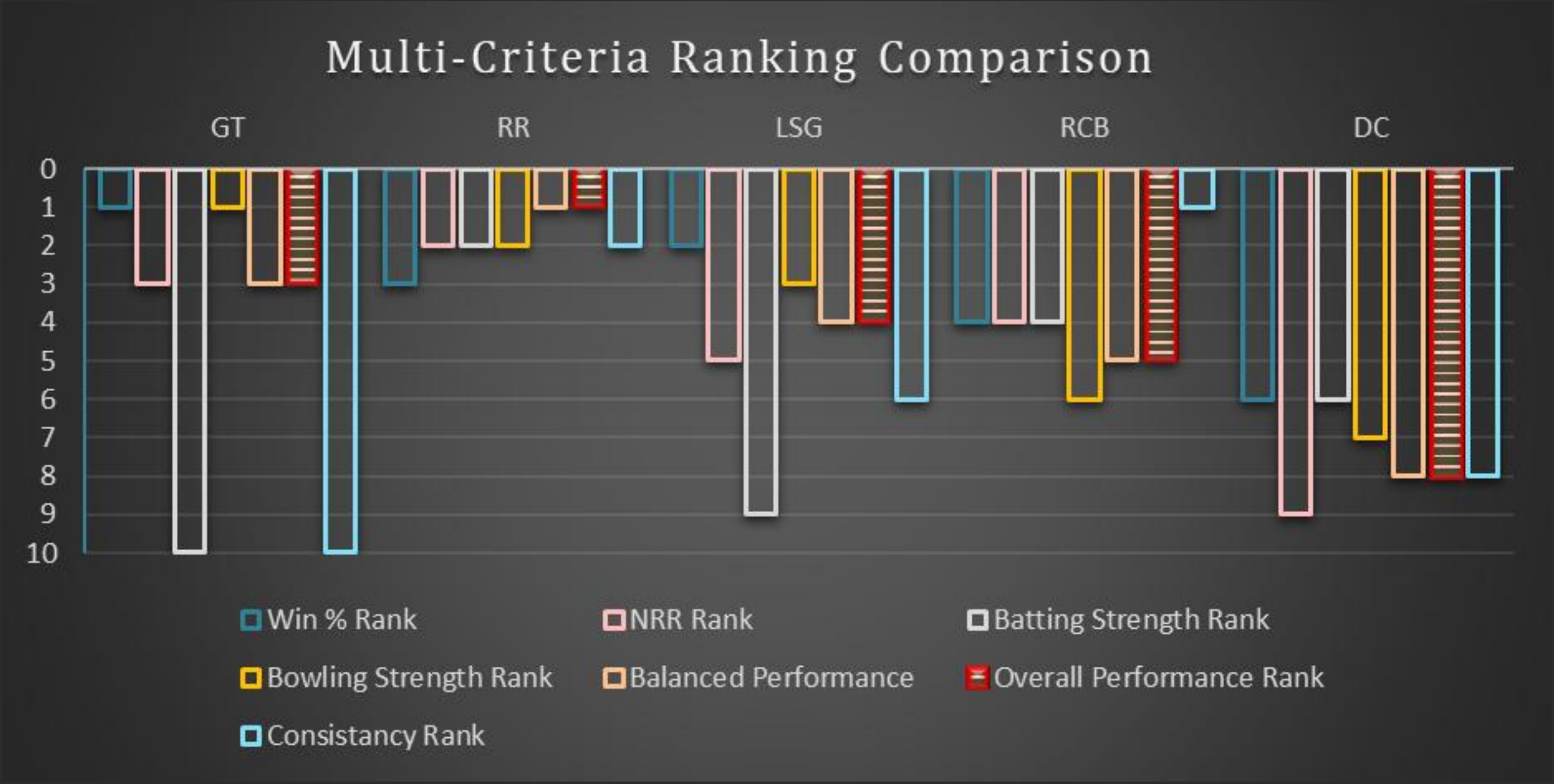
Rank	Team	Why They Rank Here
6	Punjab Kings (PBKS)	Mid-level dominance with fluctuating performance
7	Chennai Super Kings (CSK)	Inconsistent dominance across seasons
8	Delhi Capitals (DC)	Average performance with limited long-term dominance
9	Sunrisers Hyderabad (SRH)	Low dominance and inconsistent results
10	Mumbai Indians (MI)	Lowest sustained dominance across the three seasons

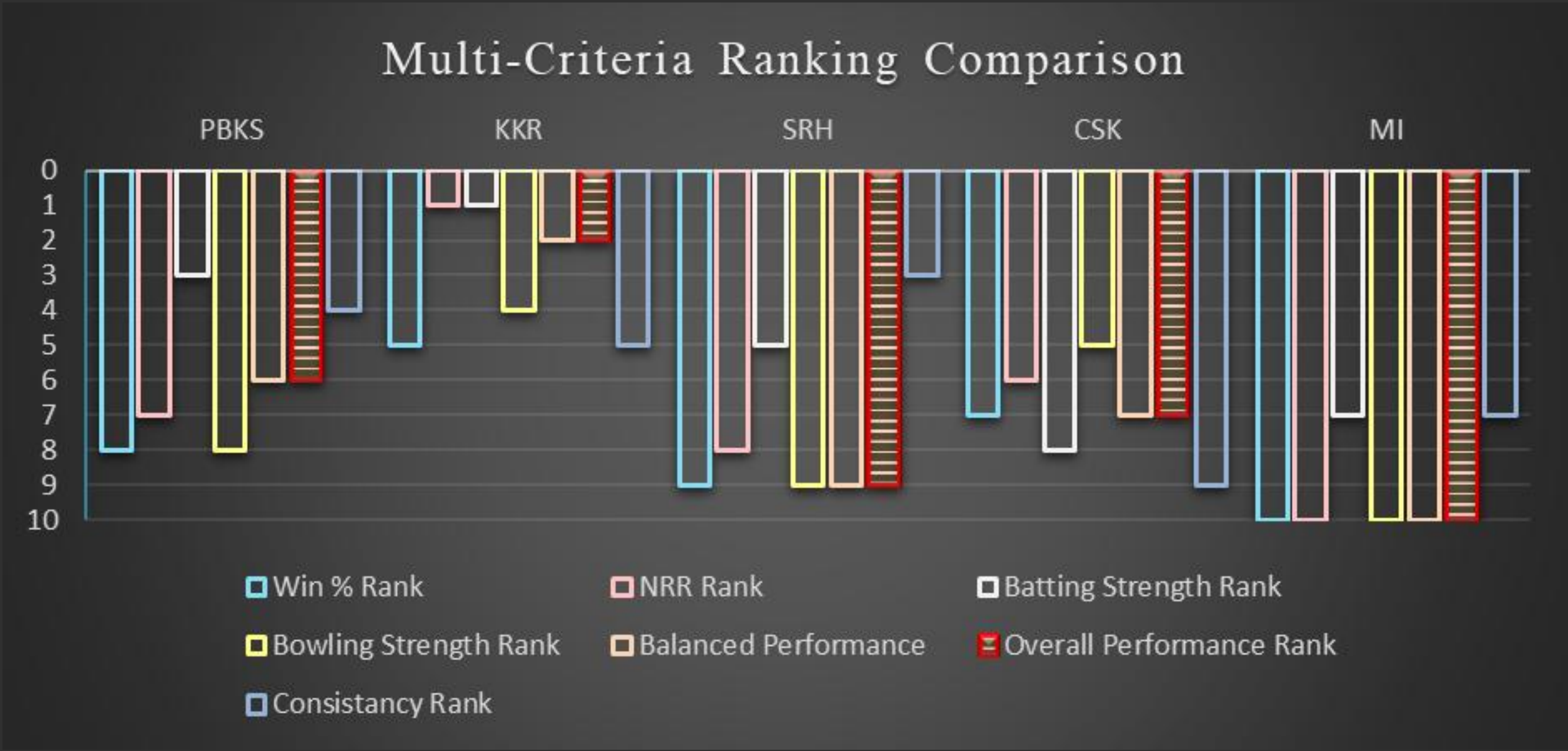

Dominance VS Consistency
120%
100%
80%
60%
40%
20%
0%
CI
30%
35%
40%
45%
50%
55%
60%
65%
70%
Average SDI
RCB
SRH
PBKS
RR
KKR
LSG
MI
DC
CSK
GT


Team Skill Balance Analysis
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
GT
RR
LSG
RCB
DC
PBKS
KKR
SRH
CSK
MI
Batting Strength
Bowling Strength
Balanced Performance


Dominance Index from Performance Indicators
80%
70%
60%
50%
40%
30%
20%
10%
0%
GT
RR
LSG
RCB
DC
PBKS
KKR
SRH
CSK
MI
Win %
NRR %
Balanced Performance
Dominance Index


Overall Rank VS Win % Rank
GT
RR
LSG
RCB
DC
PBKS
KKR
SRH
CSK
MI
0
1
2
3
4
5
6
7
8
9
10
Win % Rank
Overall Performance Rank


Multi-Criteria Ranking Comparison
GT
RR
LSG
RCB
DC
0
1
2
3
4
5
6
7
8
9
10
Win % Rank
NRR Rank
Batting Strength Rank
Bowling Strength Rank
Balanced Performance
Overall Performance Rank
Consistancy Rank


Multi-Criteria Ranking Comparison
PBKS
KKR
SRH
CSK
MI
0
1
2
3
4
5
6
7
8
9
10
Win % Rank
NRR Rank
Batting Strength Rank
Bowling Strength Rank
Balanced Performance
Overall Performance Rank
Consistancy Rank

Key Takeaways

- Balanced teams perform better in the long run
- Consistency matters more than one-season success
- Win percentage alone is not enough
- Net Run Rate influences dominance strongly
- Performance fluctuations lower final rankings
- Teams with balanced skills and consistent performance rank higher over time.
- Analytical metrics offer a clearer and more accurate evaluation

End of Presentation

Thank You